# HUGUES ROMAIN

*Freelance Full-Stack & AI Engineer — architecture, AI agents, end-to-end web & mobile products*

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## EDUCATION

**HETIC — Grande École du Web** — Montreuil, France

*Master's Degree (M.Sc. equivalent) — Computer Programming & Specific Applications* — Nov. 2021 – Sept. 2023

Work-study program alongside Alobees, then Microdoing.

*Bachelor's Degree — Computer Programming & Specific Applications* — 2018 – 2021

Work-study program alongside Ouihelp.

## PROFESSIONAL EXPERIENCE

**Freelance — Full-Stack & AI Engineer · Axiome (SASU)** — Paris · Sept. 2024 – Present

*Founded and run my own company; client engagements and in-house products, from architecture to production.*

**aiOrigin** | *Co-founder & Lead Engineer* — Feb. 2026 – Present

*Multi-tenant SaaS platform for autonomous AI agents in the enterprise.*

- Designed and built the platform end to end — architecture, fully async FastAPI backend, React 18 / TypeScript frontend, Docker infrastructure and AI layer — on PostgreSQL (Supabase), Redis and Temporal Cloud.
- Engineered a multi-provider agent loop (Claude Agent SDK + LiteLLM) making 4 LLM providers interchangeable without touching agent code, with semantic tool selection across 850+ OAuth-connected applications.
- Built token-by-token real-time streaming (Redis Streams → SSE → React): idempotent sequenced events, crash-resume and inactivity watchdog, removing perceived wait on multi-minute responses.
- Orchestrated long-running work through durable Temporal workflows — scheduled jobs and a 4-phase multi-agent B2B prospecting pipeline — with dual heartbeats and automatic cleanup of orphaned runs.
- Secured multi-tenancy: ES256 JWT auth (JWKS), PostgreSQL RLS, a 5-level permission hierarchy, sandboxed tool execution and idempotent Stripe usage-based billing.
- Set up observability (Sentry, PostHog, Langfuse over OpenTelemetry) and a distributed documentation convention (~48 linted AGENTS.md files) consumable by AI agents.

**Sinea Formation** | *Lead Full-Stack Engineer* — Sept. 2025 – Present

*B2B platform helping employees develop their soft skills.*

- Contributed to product design and scoping alongside the client: specification workshops, functional trade-offs and roadmap prioritisation.
- Led the technical side of the project: work breakdown, code reviews and coordination of the developers across the three repositories.
- Built and evolved the product's three codebases: Expo / React Native mobile app, NestJS + Prisma + PostgreSQL API and React / TanStack Query back office.
- Shipped the app to iOS and Android stores (EAS builds, native versioning, App Store & Play Store submissions) on a sustained release cadence.
- Built the campaign and challenge engine — progress tracking, badges, per-team statistics — and multi-format feedback (text, audio, video with Deepgram transcription).
- Implemented GDPR compliance (JSON export of personal data, cascading account deletion) and multi-channel notifications (Firebase, email, Teams, SMS).
- Hardened the API: 4-role JWT authentication, guards and scopes (UserScoped / ClientScoped), throttling and Sentry monitoring.

**Axiome App Studio** | *Founder & Product Engineer* — 2025 – Present

*Consumer mobile app studio, designed and operated solo.*

- Designed, built and shipped 5 mobile apps (Expo SDK 56/57, React Native, strict TypeScript, Supabase, Superwall): Belle (AI beauty advisor), Koji (gut health), Tanzy (tanning coach), KnuckleMax and Teacherz (AI tutoring chat).
- Put AI into production on the product side: vision-model image analysis, personalised onboarding paths, generated content and auto-detected FR/EN/ES/IT localisation.
- Industrialised the full release chain — multi-file version bumping, local .ipa / .aab builds, App Privacy, App Review responses — reducing each submission to an automated procedure.
- Instrumented the whole portfolio with PostHog: tracking plan, onboarding funnels, activation and retention monitoring, driving product iterations from data.
- Set up subscription monetisation (Superwall paywalls) and Sentry error tracking across every app.

**Ida** | *Full-Stack Engineer* — Sept. 2024 – Sept. 2025

*B2B ordering platform for fresh produce serving neighbourhood grocery stores.*

– Designed and built the core ordering and fresh-product management features of the platform (FastAPI, React).
– Delivered a dynamic price list: real-time pricing, packaging units, product groups and business rules reacting to stock movements.
– Instrumented the product with PostHog to track user behaviour, prioritise the roadmap and optimise key flows.
– Coordinated the tech team day to day (code reviews, agile rituals) to smooth collaboration and speed up delivery.

**Microdoing** Paris · Oct. 2022 – Oct. 2023

*Full-Stack Engineer — work-study*

– Designed and built a gamification feature that lifted user engagement and adoption by 20%.
– Ported the React Native mobile app to the web with Expo for Web, opening a new access channel to the product.
– Reorganised the team (code reviews, sprints, daily rituals, retrospectives) to cut the production error rate.

**Alobees** Paris · Dec. 2021 – Sept. 2022

*Full-Stack Engineer — work-study*

– Rearchitected the web and mobile front ends and unified both around an atomic design system.
– Built an interactive site-planning tool letting construction professionals visualise, organise and track their crews.
– Removed UX friction from sign-up and login flows, from design through to implementation.
– Introduced automated testing (Jest), strengthening code robustness and maintainability.

**Ouihelp** Paris · Jul. 2019 – Sept. 2021

*Front-End Engineer — work-study*

– Designed and built a smart location feature to quickly identify the caregivers closest to each beneficiary.
– Reworked state management (replacing Redux with React hooks and contexts), improving app performance.
– Built a client portal simplifying access to Ouihelp services for families and beneficiaries.
– Created a Gatsby static site wired to Contentful (headless CMS) for autonomous content management.

## TECHNICAL SKILLS

**Languages —** TypeScript, JavaScript, Python, SQL

**Front-end —** React 18, Next.js, React Native / Expo, TanStack Query, Zustand, Tailwind CSS, Radix UI, design systems, i18n

**Back-end —** FastAPI, NestJS, Node.js, PostgreSQL / Supabase, Prisma, SQLModel, Redis, pgvector, REST & SSE

**AI & agents —** Claude Agent SDK, MCP, LiteLLM (Anthropic · OpenAI · Gemini · Mistral), RAG, multi-agent orchestration, Temporal

**Infra & tooling —** Docker, Temporal Cloud, Stripe, Railway, EAS / Expo, GitHub Actions, Sentry, PostHog, Langfuse

**Product —** Software architecture, App Store & Play Store publishing, GDPR compliance, team leadership, agile methods

## PROJECTS & ADDITIONAL

**Projects —** Astrotierra — astrocartography web product designed and launched solo (Supabase Edge Functions, Swiss Ephemeris, LLM-generated readings). Signal Radar — SaaS detecting app opportunities from trend signals.

**Languages —** French (native) · English (fluent)

**Code —** github.com/huguesromain